Nautilus

AI-Enabled Education and Research on the National Research Platform

Overview, Jupyter and Jupyter AI

NAIRR Pilot Annual Meeting

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San Diego Supercomputer Center

NRP NATIONAL RESEARCH PLATFORM

SAN DIEGO SUPERCOMPUTER CENTER

UC San Diego

Agenda

- Overview, JupyterAI, and Hosted JupyterHub Integration
- Requesting GPUs on NRP *(Morning break during this talk)*
- Inference Workloads using NRP resources including NVIDIA A10 GPUs and Qualcomm Cloud AI 100 Ultra Cards with Hugging Face Models

Lunch

- Multi-Tenant Vector Databasing with Milvus for RAG and Fine-tuning with GPU/QAIC
- Custom Hosted JupyterHubs *(Afternoon break during this talk)*
- Agentic Workflows with Coder and “Vibe Coding

GitHub Repository for Today's Tutorials

https://github.com/nrp-nautilus/nairr-tutorial/tree/main

Launch Tutorial Workspace

- We are going use a Jupyter workspace for all of today's hands-on content. Get started by clicking on the link in the github page
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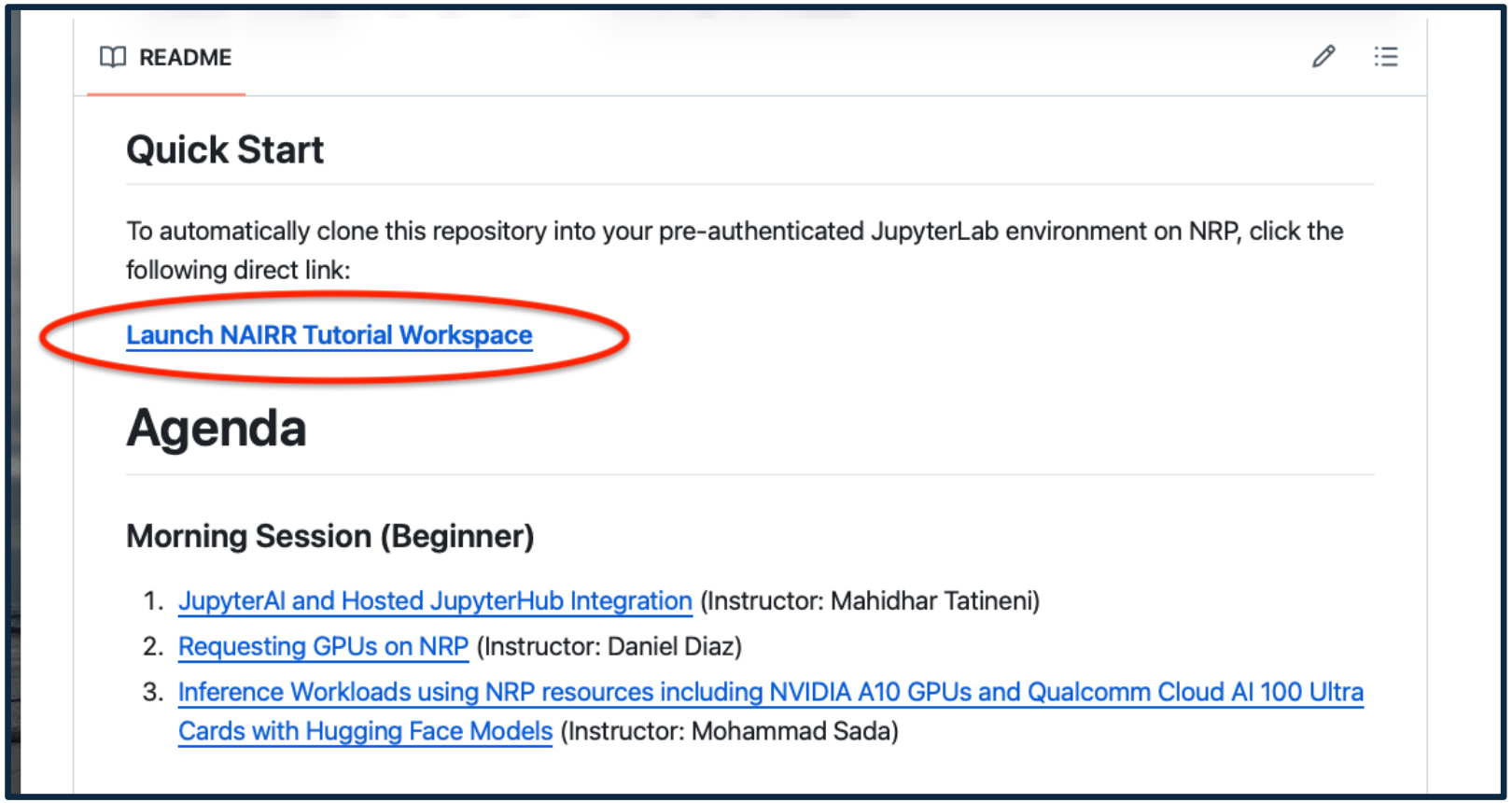

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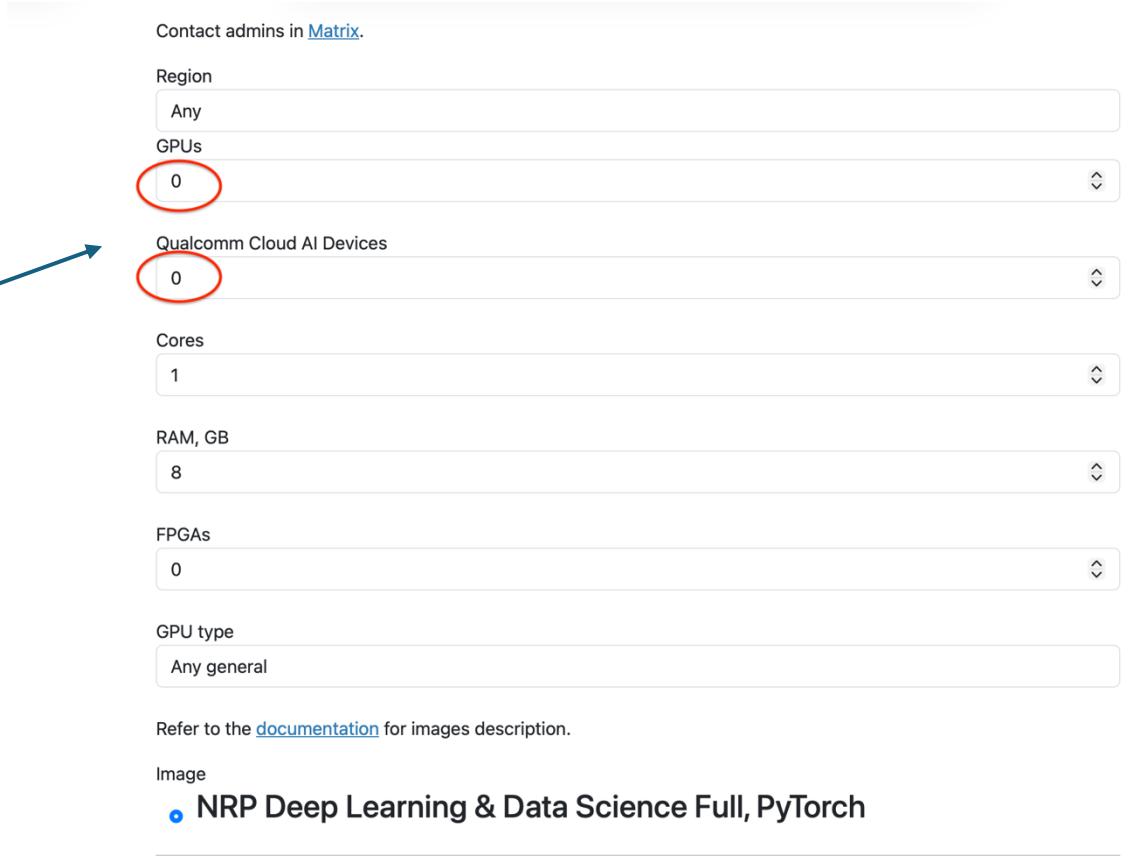
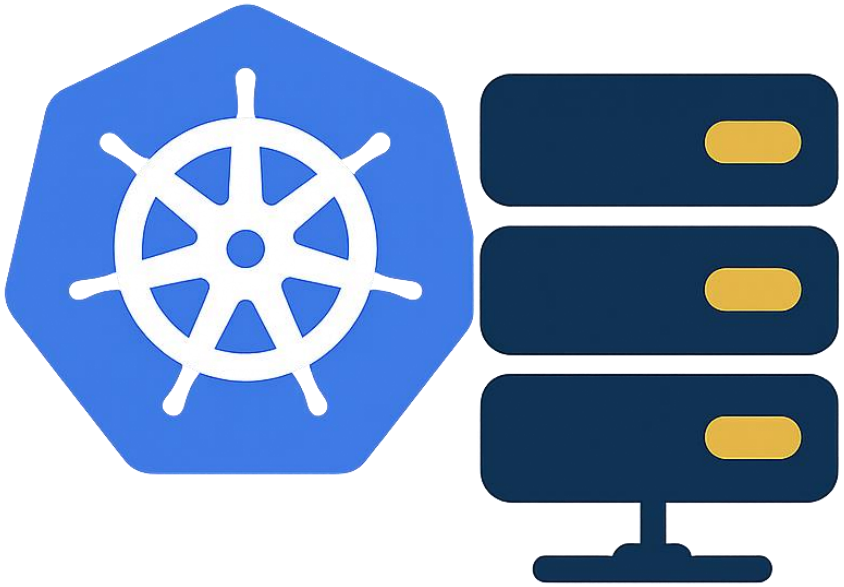




Introduction to the National Research Platform

NAIRR Pilot Workshop

Motivation: Current Challenges & Outlook

Current Challenges

- **Rising Demand:** Growing reliance on cyberinfrastructure (CI) for research and education.
- **Access Inequality:** Widening gap between institutions with and without CI resources
- **Architectural Fragmentation:** Post-Moore's Law era complicates domain science adoption.

Future Outlook

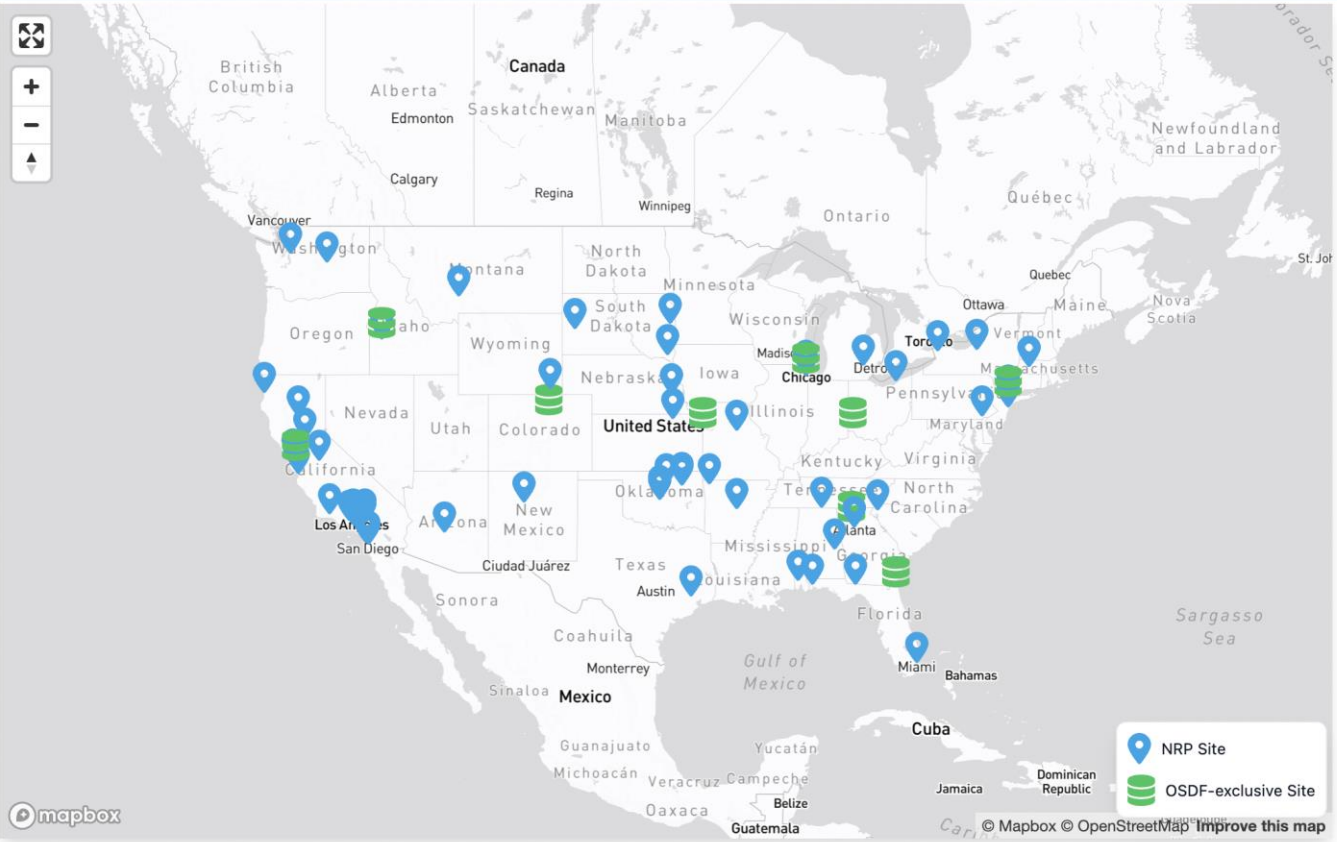
- **Equity Focus:** Prioritize CI access for *all* institutions.
- **Universal Connectivity:** Connect educational institutions to advanced CI.
- **Talent Empowerment:** Build environments for collaboration.
- **Global Leadership:** Invest holistically in CI to drive innovation and competitiveness.

The National Research Platform

- The National Research Platform is a partnership of more than **80** institutions.
- Led by researchers at **UC San Diego, University of Nebraska-Lincoln, and MGHPCC.**
- Includes contributions by the **National Science Foundation**, the D**epartment of Energy**, the **Department of Defense**, and many research universities and R&E networking organizations in the US and around the world.
- NRP operates a primary **Kubernetes** distribution cluster with hardware spanning across **3 continents**, called **Nautilus**.

National Research Platform

Docs



NRP NATIONAL RESEARCH PLATFORM

The National Research Platform is a partnership of more than 50 institutions, led by researchers at UC San Diego, University of Nebraska-Lincoln, and Massachusetts Green High Performance Computing Center and includes contributions by the National Science Foundation, the Department of Energy, the Department of Defense, and many research universities and R&E networking organizations in the US and around the world.

Select a site or click on a site in the map

Select...

Sites	Nodes
84	**439**
Sites hosting NRP nodes	Nodes registered in Kubernetes
GPUs	**CPU Cores**
1,496	**29,254**
Total GPUs across all nodes	Total CPU cores across all nodes

GPUs Allocated

944 Sep 2, 2025, 10 AM

Sep 1, 2025, 10 AM — Sep 2, 2025, 10 AM

Running Jupyter Pods

49 Sep 2, 2025, 10 AM

Sep 1, 2025, 10 AM — Sep 2, 2025, 10 AM

Active Research Groups using GPUs

90 Sep 2, 2025, 10 AM

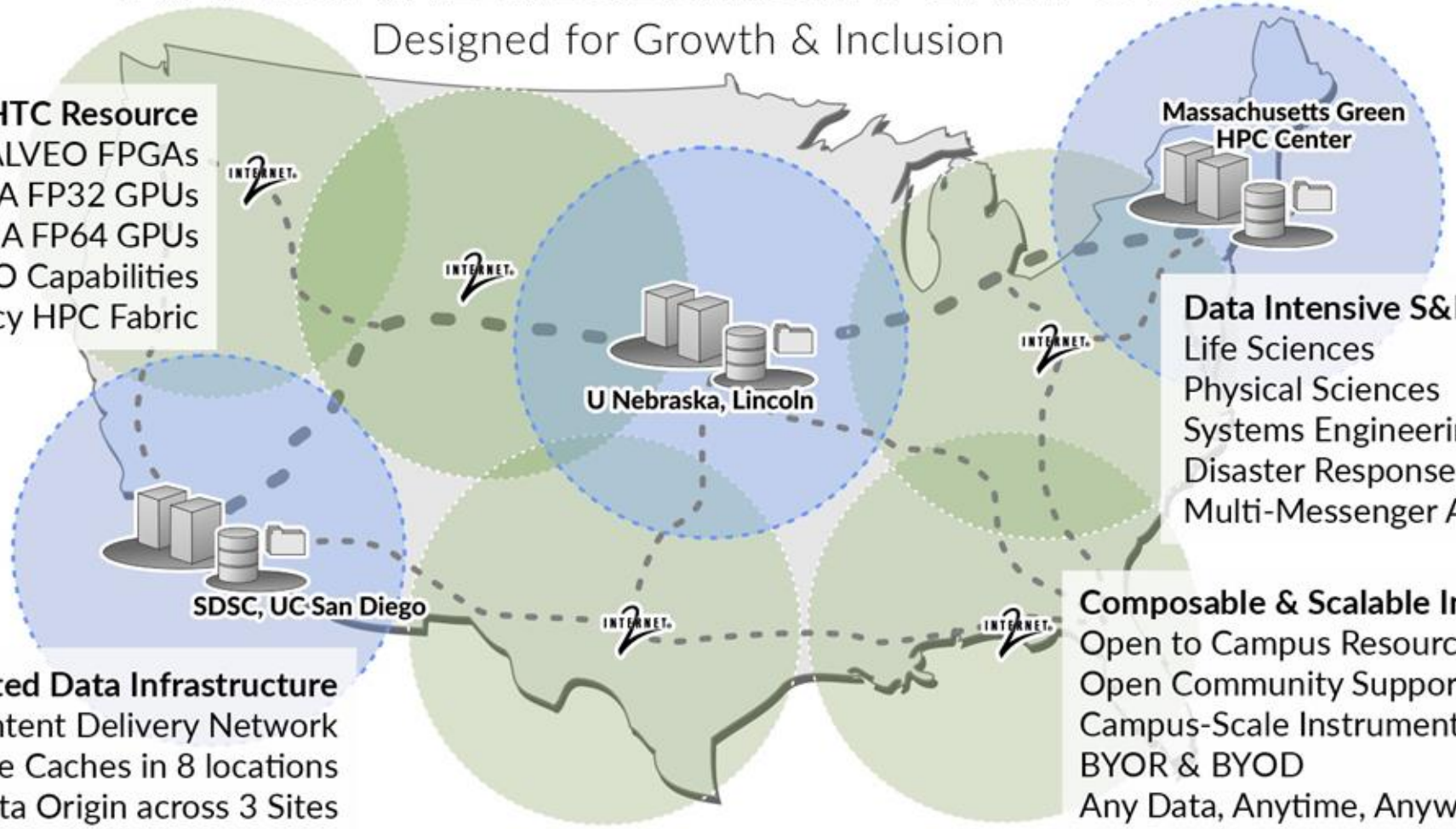
Sep 1, 2025, 10 AM — Sep 2, 2025, 10 AM

https://dash.nrp-nautilus.io/

NATIONAL RESEARCH PLATFORM

Designed for Growth & Inclusion

HPC/HTC Resource
32 ALVEO FPGAs
A10 288 NVIDIA FP32 GPUs
80GB A100 64 NVIDIA FP64 GPUs
Tbps WAN IO Capabilities
GigaIO's Low Latency HPC Fabric



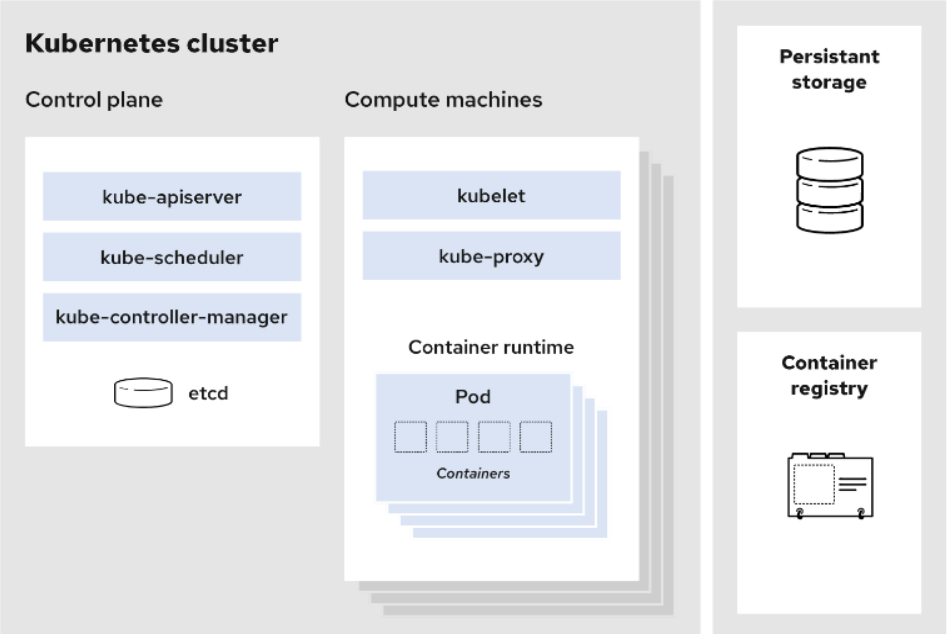
Data Intensive S&E
Life Sciences
Physical Sciences
Systems Engineering
Disaster Response
Multi-Messenger Astrophysics

Composable & Scalable Innovation
Open to Campus Resource Integration
Open Community Support Model
Campus-Scale Instrument integration
BYOR & BYOD
Any Data, Anytime, Anywhere

Distributed Data Infrastructure
National Scale Content Delivery Network
50TB 100Gbps NVMe Caches in 8 locations
4.5PB Distributed Data Origin across 3 Sites

The Nautilus Cluster

- Nautilus is a Kubernetes cluster
- Control plane: Manages the cluster and the worker nodes
- Worker Nodes: Maintain running pods and provide Kubernetes runtime environment
- Has been in continuous operation for **6 years**.
- Various storage options like **CephFS**; **CvmFS**; **S3**; advanced monitoring tools like **PerfSONAR**; **traceroute**; **Prometheus**.
- Diverse computation and data tools: **JupyterHub**; **WebODM**; **GitLab**; **Nextcloud**; **Overleaf**.
- Collaboration and communication: **Jitsi**; **EtherPad**; **Hedgedoc**; **SyncThing**.
- **430+ nodes**; **1400+ GPUs**; **50+ FPGAs**; **Bluefield 2 SmartNICs**; **SmartSSDs**; programmable **Tofino** switches....

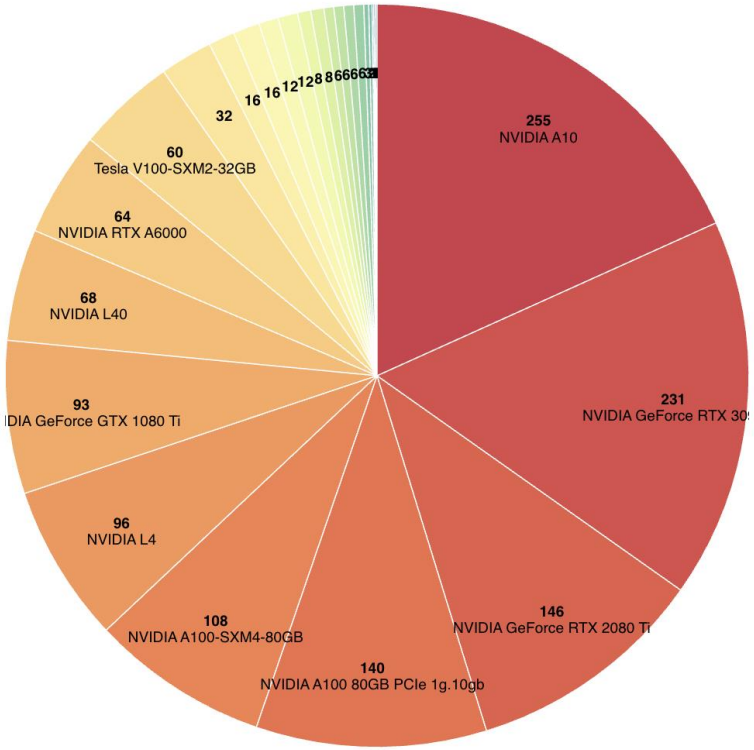


Hardware on Nautilus

- **ARM + x86_64 Nodes**
- **1400+ GPUs with various models**
  - Good for training most of the ML models
  - 4-8 GPUs / node: can request all of them
  - 550 GB RAM

~600 RTX/GTX "Gaming" GPUs

- **8 Qualcomm Cloud AI 100 Ultra Inference & Fine-tuning Cards:**
  - 1 card = 4 SoCs (System-on-Chip)
  - Each SoC capable of running 25B Param LLM with out-of-the-box config
  - Provisioned via k8s device plugin



https://nrp.ai/viz/

https://grafana.nrp-nautilus.io/

Hardware on Nautilus

Cluster Resource Summary

The cluster has different types of resources including CPU, GPU,and FPGAs.

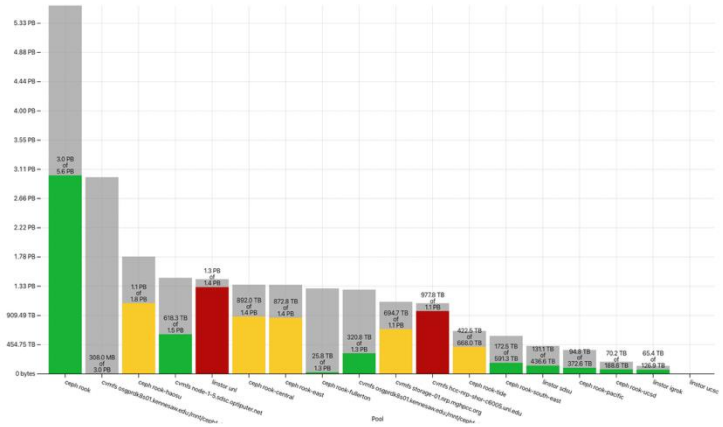
- **CPUs:** 16 - 384 cores / node
- **GPUs:** max 16 in a node
- **FPGAs:** max 4 in a node
- **Memory (RAM):** 16 GB - 1.6 TB / node
- **Disk Space (local scratch):** 107.3 GB - 17.8 TB / node

https://nrp.ai/viz/resources

- ConnectX-6 NICs, P4-programmable **Tofino** Switches
- **32** P4-programmable Xilinx Alveo U55C **FPGAs**
- **16** BittWare 520N-MX Stratix **FPGAs**
- **3** Xilinx Alveo SN1000 **FPGAs**
- **7** NVIDIA Bluefield 2 **DPUs**
- **2** Xilinx U.2 **FPGA-powered SmartSSDs**
- **Kubernetes** cluster with ***Multus, Calico and ESnet SENSE support.***
- *+ Agilex7-M FPGAs + Various Types of Hardware*

Storage Options on Nautilus

- Ceph File System → Ceph FS → ReadWriteMany PersistentVolumes
- Ceph Rados Block Device → Ceph RBD → ReadWriteOnly PersistentVolumes
- Linstor Block → ReadWriteOnly PersistentVolumes
- S3 Compatible storage → Bucket Endpoints
- Open Science Data Federation → OSDF



Using Nautilus and NRP

JupyterHub

Coder

Kubernetes

GitHub Repository for Today’s Tutorials

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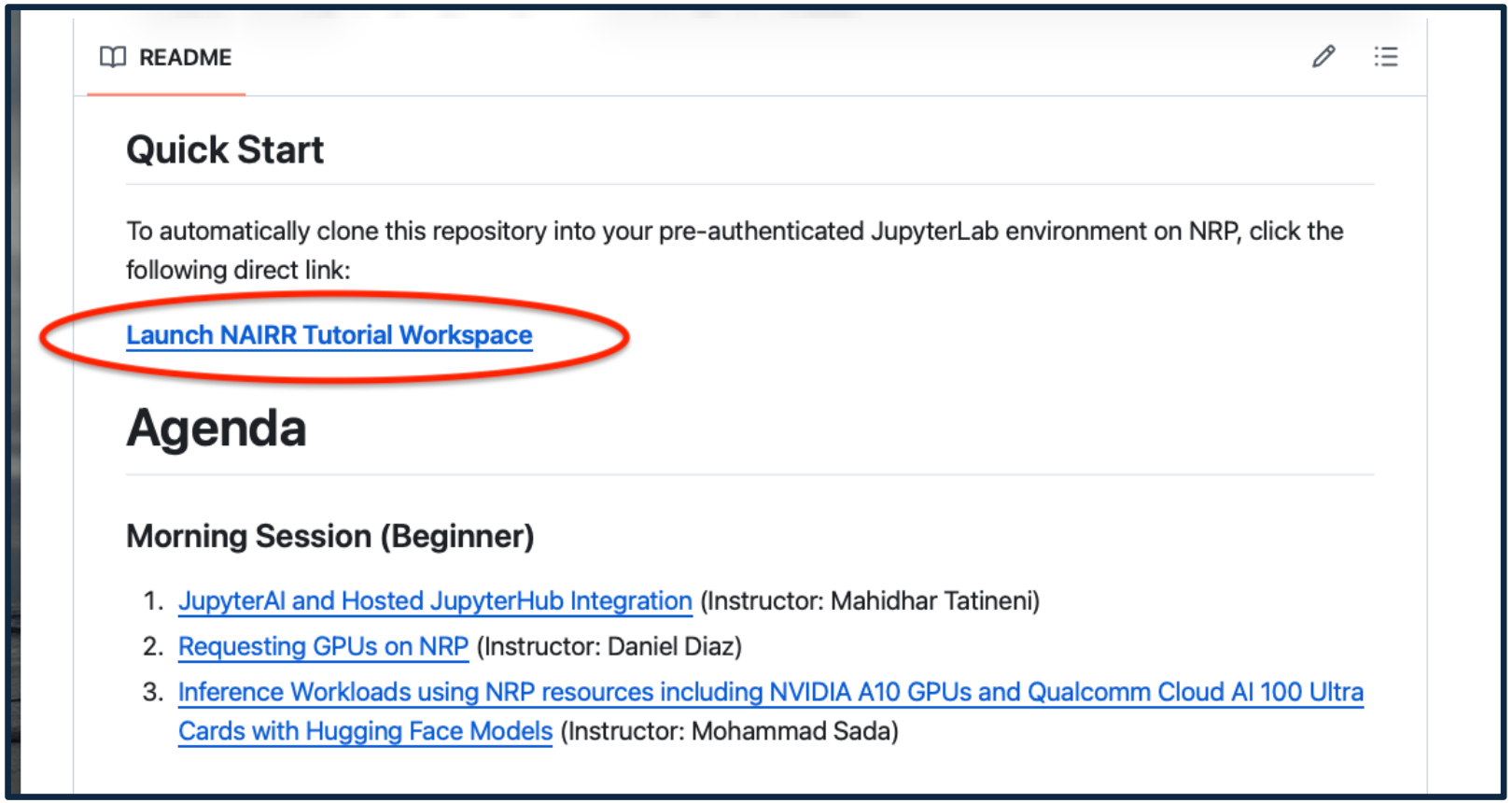

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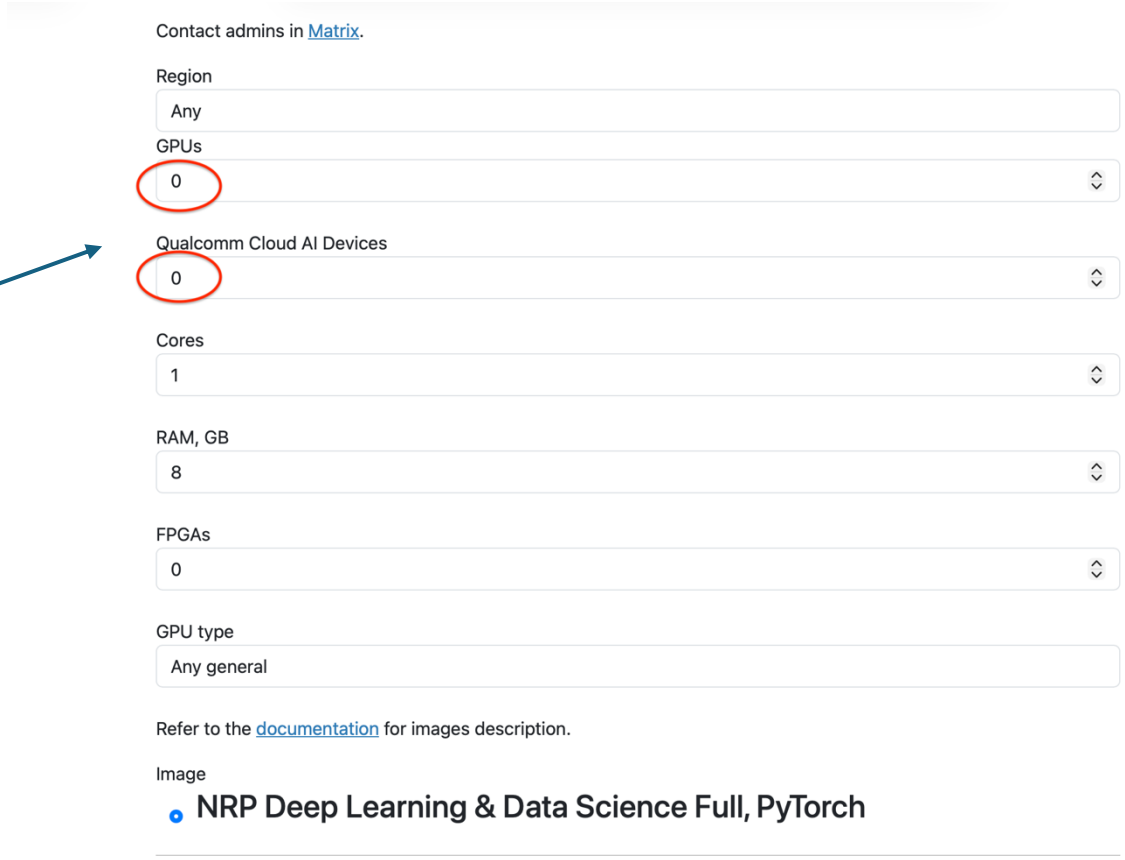
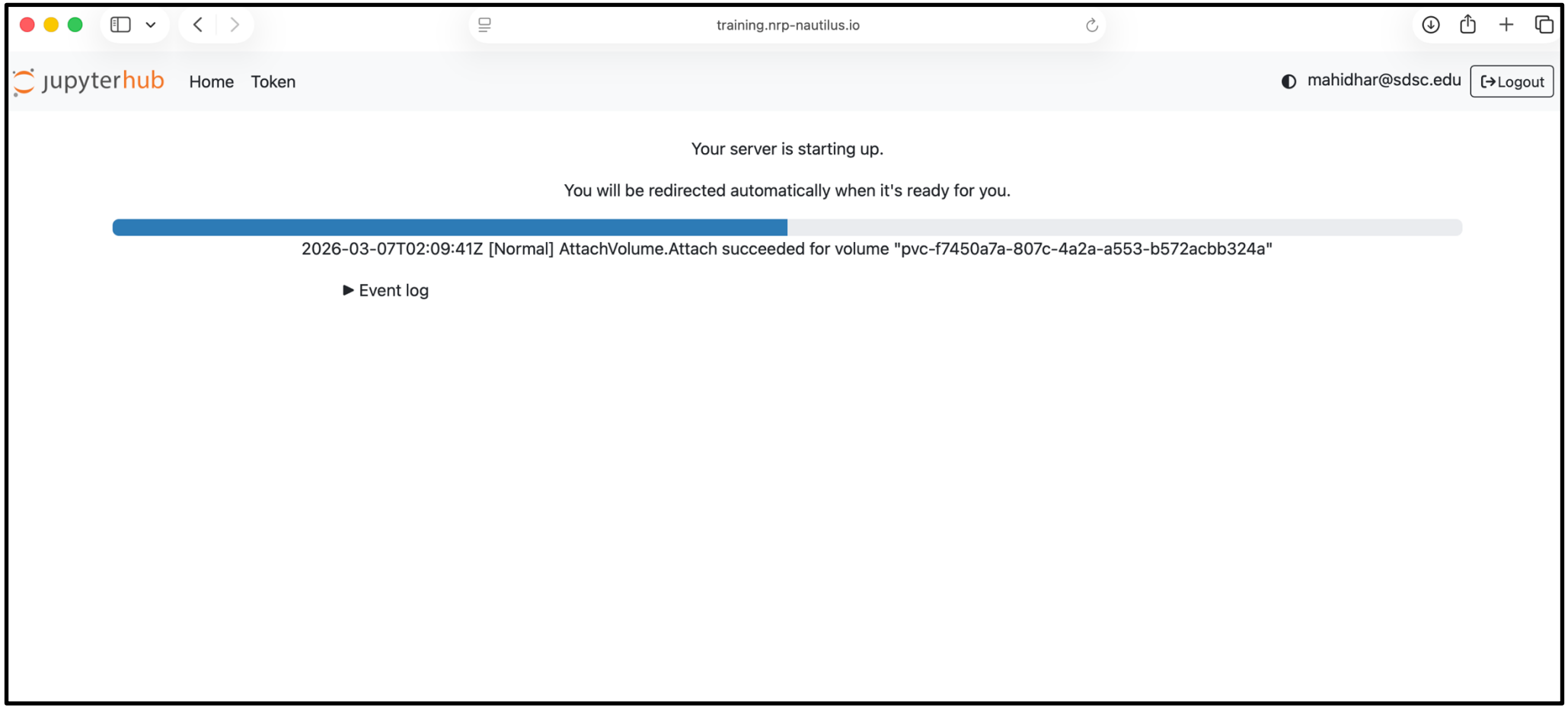


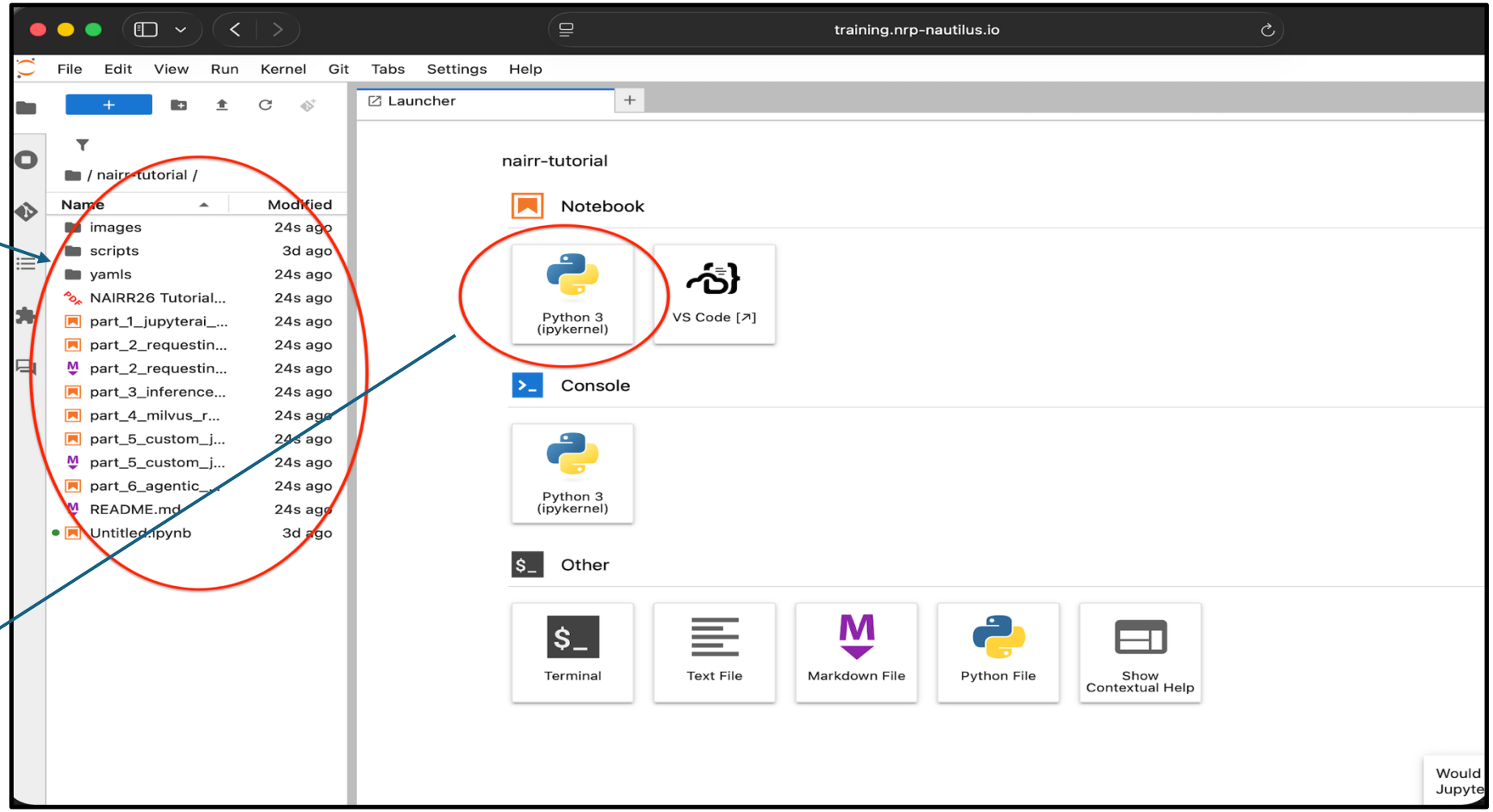
Jupyter server startup process



Jupyter instance with clone of our repository

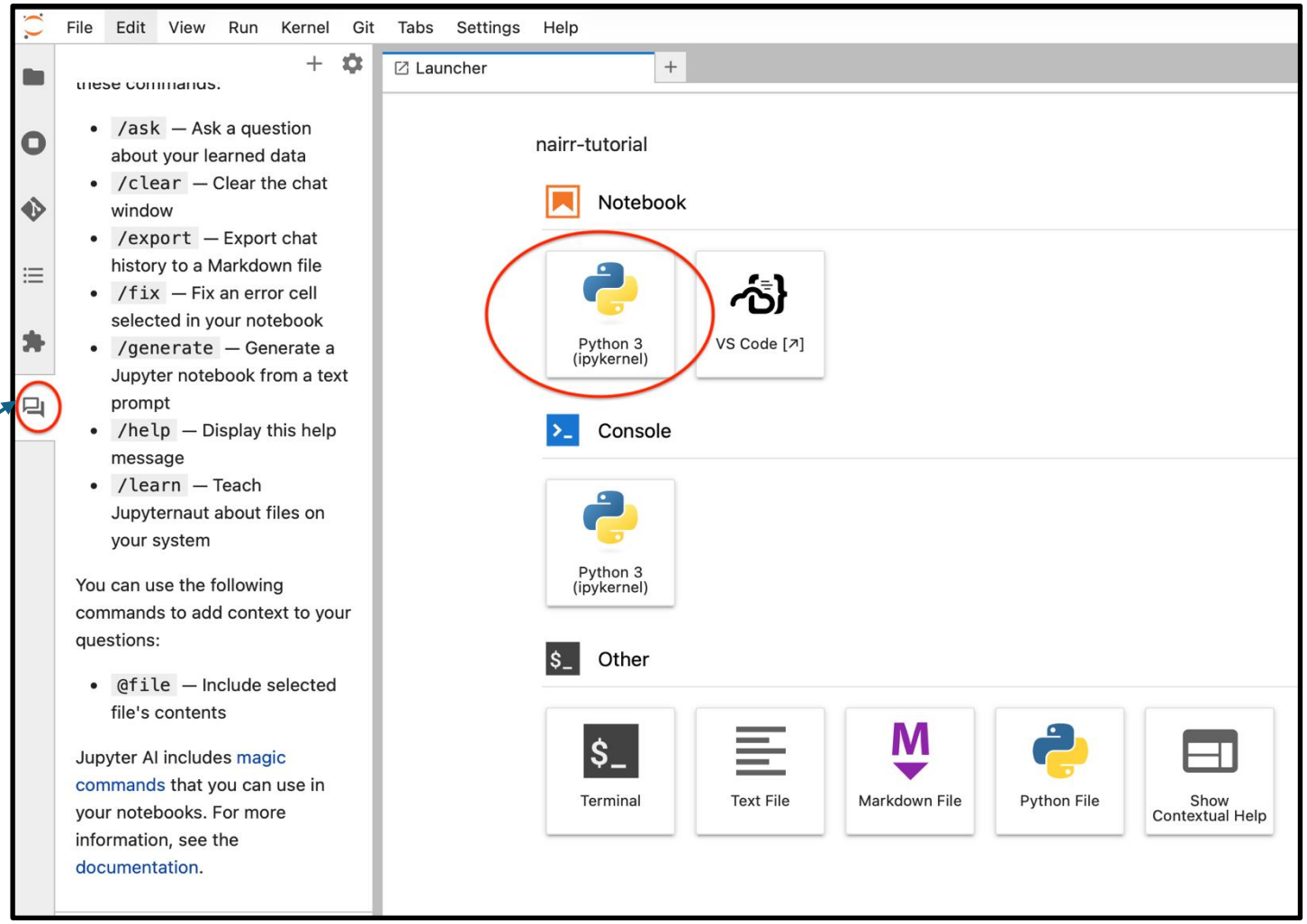
Clone of our git repo

Click on Python3 notebook icon so we can test things out. Later examples will use the notebooks on the left.



Jupyter AI Chat interface

Click here for chat interface



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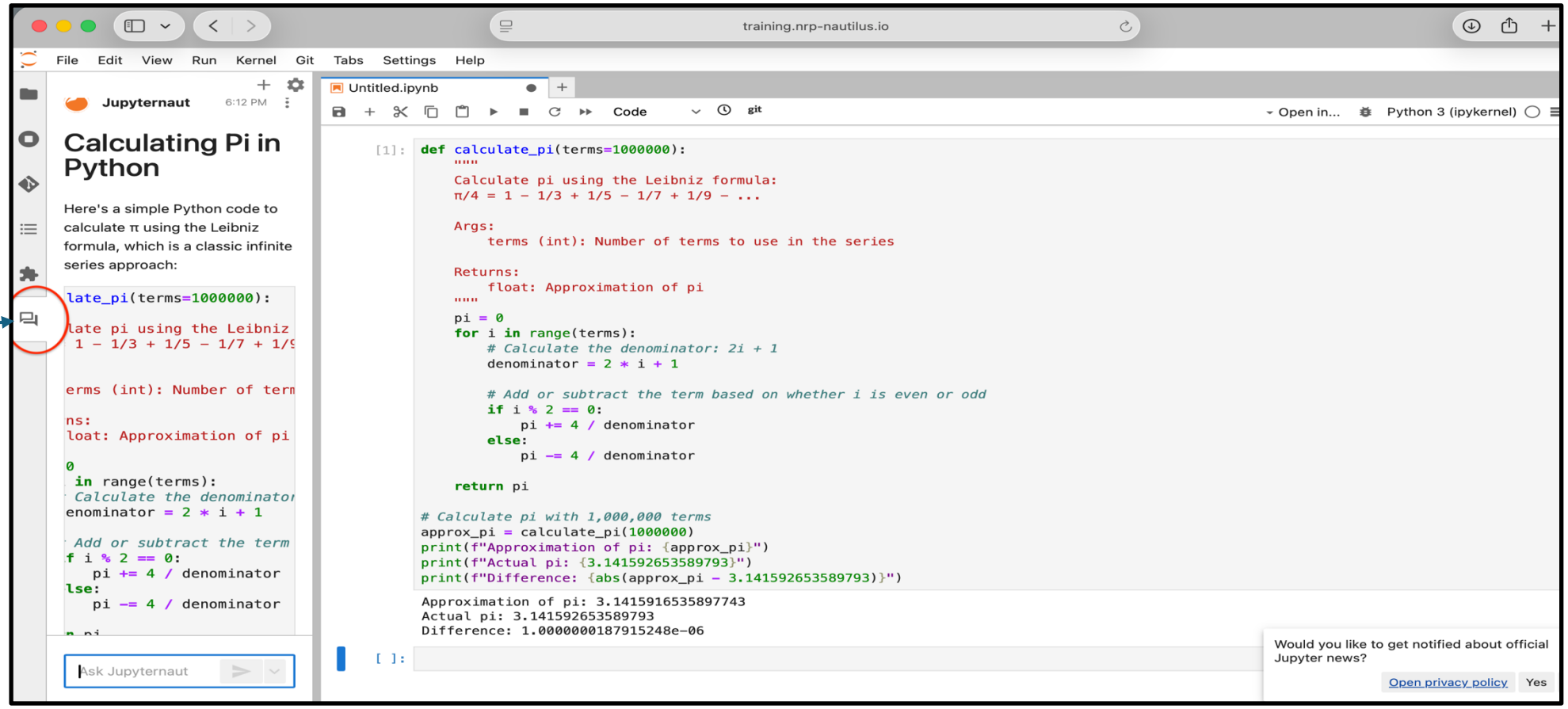
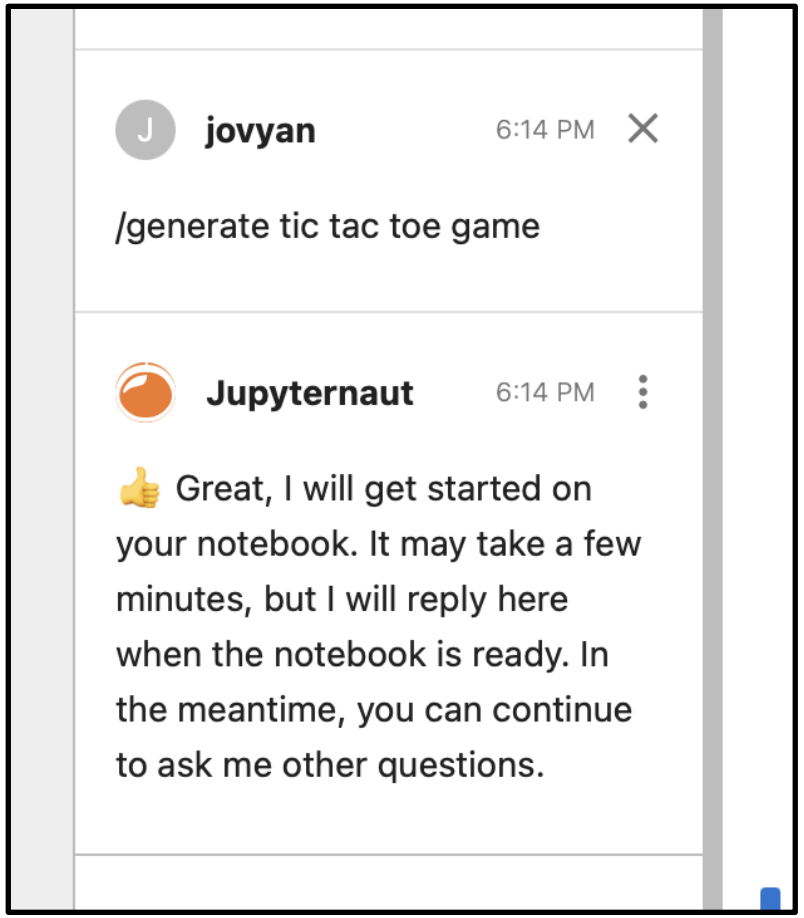
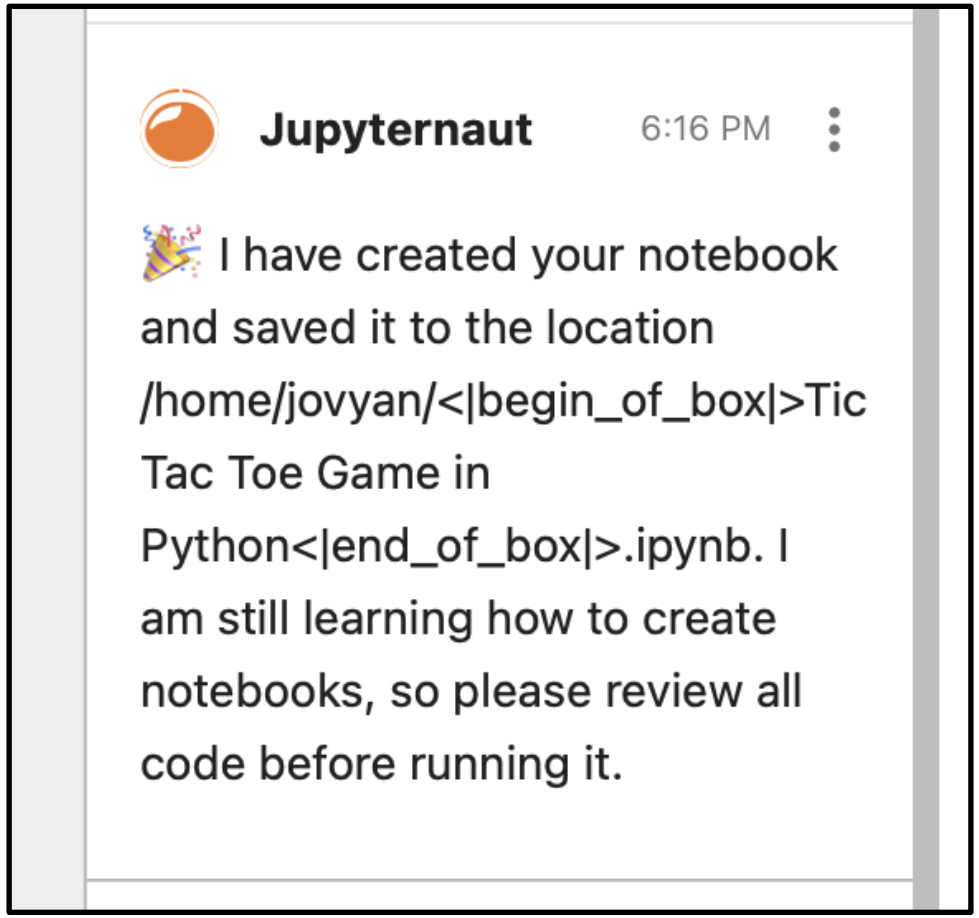


Jupyter AI Chat interface – Generate notebooks

Request in Chat Interface



After a few minutes … a notebook!



Jupyter AI – Cell Magic!

```
[15]: %reload_ext jupyter_ai_magics

[18]: %%ai openai:gemma3
      Show a code to do bubble sort

[18]:
def bubble_sort(list_):
    """
    Sorts a list using the bubble sort algorithm.

    Args:
    list_: The list to be sorted.

    Returns:
    The sorted list.
    """
    n = len(list_)
    for i in range(n):
        # Flag to optimize - if no swaps occur in a pass, the list is sorted
        swapped = False
        for j in range(0, n-i-1):
            # Compare adjacent elements
            if list_[j] > list_[j+1]:
                # Swap if the element found is greater than the next element
                list_[j], list_[j+1] = list_[j+1], list_[j]
                swapped = True
        # If no two elements were swapped in inner loop, the array is sorted
        if not swapped:
            break
    return list_

# Example usage
my_list = [64, 34, 25, 12, 22, 11, 90]
sorted_list = bubble_sort(my_list)
```

Commands used in the cells:

%reload_ext jupyter_ai_magics

%%ai openai:gemma3
Show a code to do bubble sort

NAIRR National Artificial Intelligence Research Resource

2026 ANNUAL MEETING
MARCH 10-13, 2026

Tutorial Survey

https://bit.ly/NAIRR26TutorialSurvey

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